# Reply to reports on article JBES-P-2022-0567.R1

First of all, we would like to once more thank the editor, the associate editor and the reviewers for their thoughtful comments and suggestions which have greatly helped us improve the paper. In this revision, we have carefully addressed all points raised and made corresponding changes to the paper accordingly. Below, we provide point-by-point responses indicating where alterations to the manuscript have been made. While we incorporated as much new material as possible into the main text, some content (particularly from the revised simulation study) has still been allotted to the Supplement to meet the 35-page limit. We are happy to adjust the placement of content further if needed.

## Reply to the associate editor

1. *My apologies, but I maintain that your $H_0$ (page 3) is unnatural and uninteresting. In the trend analysis of multiple time series, it is extremely unlikely that the natural 'position of ignorance' is that all trend curves are the same. What if, for example, I already know or suspect that the curves form distinct groups? It is then something of a wasted effort to start from the position that they do not differ at all. Also, the null hypothesis is not robust to high-dimensionality as it is more likely that some curves will differ as I observe a higher number of variates (I understand that you do not do this in the paper but this only makes it even less interesting, in my view). As a cartoon illustration of the first point above, if I suspect that $p-1$ (say) curves may be the same, but 1 is distinctly different, do I need to first remove the offending one before performing your test, in which case I enter the territory of post-selection inference (which you do not cover), or do I not need to remove the offending one, in which case I am pretending that my $H_0$ is different from what it is in reality?*

   We appreciate your perspective regarding the null hypothesis $H_0$. We recognize that assuming identical trends across time series can seem restrictive in many real-world applications. However, we argue that our choice of $H_0$ is methodologically justified because it represents the simplest, most parsimonious starting point for testing. This aligns with the traditional Neyman-Pearson framework where the null serves as a benchmark for detecting departures. Furthermore, our multiscale method enhances this approach by not only rejecting $H_0$ but also identifying specific intervals and pairs of series where differences occur, providing actionable insights beyond a binary rejection decision.

   We also acknowledge concerns about the robustness of our test to high dimensionality of the covariates. While our paper does not explicitly address highdimensional asymptotics, the core methodology could be extended to incorporate sparsity-driven approaches in the future.

Regarding the cartoon illustration, suppose it is indeed true that 1 curve is distinctly different from the other $p-1$ curves. First, in this case, the rest $p-1$ curves may be different among themselves as well with some having more and some having less in common with the first one. Second, even if all $p-1$ curves are exactly the same, it very well may be that the difference between them and the first curve occurs only on some subinterval of $[0,1]$. In both these cases, our test would be able to pinpoint this difference, and the subsequent clustering will not neccesarily result in a separate group for the first, offending curve. Hence, even in this cartoon example, we would still advise applying the statistical test for all $p$ time series at the same time.

2. *The clustering part is nice and perhaps it could form the main thread of a much shorter paper.*

   We have taken your suggestion into account and put more focus on the clustering part in the current paper. (Marina: I have not done this.)

3. *The empirical application is interesting, but this is thanks to the use of clustering rather than as a consequence of testing - the null is obviously inappropriate, as Figure 6 suggests, so there is no point in assuming it only to surely reject it. In other words, "let us test our null of equality between all trends" doesn't seem a scientifically valid suggestion for this dataset. The suggestion of clustering the countries is fine, though.*

   We appreciate your point of view about the empirical application, and we would like to stress once more that the null of equality between the trends is just a starting point of our analysis. Many econometric procedures, including cointegration tests and structural break tests, involve null hypotheses that are often rejected but remain essential for empirical analysis.

   We acknowledge that in this particular example clustering may be more interesting than simply testing the global null. However, our testing framework is not just a preliminary step but an integral part of the analysis. Specifically, our clustering method is directly informed by the results of the multiscale test: the distance measure that we use for applying the hierarchical clustering is exactly the pairwise test statistics that was used in computation of the global test statistics.

   Moreover, our method can detect local trend similarities that might be missed in global clustering analysis. For example, our test does not detect any significant differences between Belgium and any other country except Netherlands in the second half of the observed interval, a statement that can not be made neither from the visual inspection of Figure 6, nor from clustering alone.

4. *Can your model (1.1) include autoregression? Either way, it would be good to see it spelled out explicitly.*

   While the current formulation does not explicitly include autoregressive terms (due to the restriction on the relationship between the covariates and the errors introduced in Assumption (C7)), autoregressive dynamics can be incorporated through the error term $\varepsilon_{it}$. We briefly discuss this on p. 7 and in Remark 2.1 (p. 8-9). Moreover, all the size and power simulations are performed including the autoregressive structure in the error terms (see p. 1 in the Supplement). (Marina: should we say it explicitly somewhere in the text?)

5. *I was unconvinced by the subtraction of the overall level on p. 6. Firstly, this clearly risks eliminating useful information (what if some curves differ from others in the overall level)? Secondly, I am not sure if it is meaningful to look at averages of (e.g. increasing) trends - for example, what is the interpretation of a "long-term average house price level in [a] country"? In "Time Series 101", we teach students not to take averages of non-stationary time series!*

   The subtraction of the overall level is essential for identification. By imposing the normalization condition $\int_0^1 m_i(u)du = 0$ for all $i$, we center trends around zero, ensuring clear separating of fixed effects and trend functions. This approach focuses the analysis on dynamic variations (e.g., changes in slope, curvature) rather than static level differences. Any level differences are still captured through the fixed effects $\alpha_i$

   As for taking the averages of non-stationary time series, we believe that this is not a problem in our method. In our model, non-stationarity in the time series is introduced through the deterministic trend functions rather than through the random covariates or the error process. Specifically, Assumptions (C1) and (C4) assure that the stochastic components of the model are stationary. Since $m_i$ is a deterministic smooth function, this operation does not suffer from the same issues as averaging non-stationary stochastic processes.

   That said, it is still possible to perform the test without subtracting the overall level. In this case, the critical values will need to be calculated in a similar manner, without subtracting the averages. The rest of the testing procedure will remain the same. This approach will allow the researchers to compare persistent level shifts across the trend functions (which can potentially lead to better interpretability) as well as dynamic trend differences.

6. *(C5) - you use the delta notation before defining it. Why do your covariates X have to be random, e.g. why can't you have one of them as (say) a linear trend?*

   Thank you for pointing out the premature use of delta notation. We have now revised the manuscript to define it clearly before use.

   Regarding covariates, the assumption of random $\mathbf{X}_{it}$ is critical for identification. Specifically, if we include a linear (or any other smooth) trend as one of the covariates, it would create ambiguity about whether the observed time variation arises from the trend function $m_i$ or it is the effect of this covariate. Such ambiguity leads to non-identifiability of the deterministic trend function. Random covariates provide exogenous variation that helps disentangle their effects from those of the nonparametric trend $m_i$.

7. *Your comment (iii) on p. 9. But isn't it the case that for near-unit root processes, you will have a variety of different scenarios (some stochastic trends pointing strongly upwards, some downwards, some with no specific direction) - a situation that should be easy to discriminate from your typical scenarios of interest?*

8. *P. 11. "Throughout the paper, we assume that the long-run error variance is the same for all i, (...) [but we] use a different estimator of $\sigma^2$ for each i - this seems suboptimal.*

   (Marina: Should we rerun everything with the averages of the long-run error variances? I think I tried this and the results were somewhat similar.)

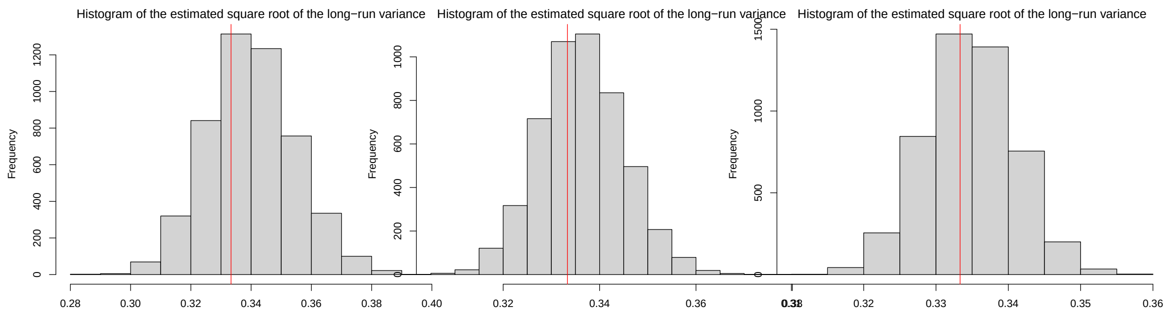
9. *I was disappointed to see that the method was not able to handle missing values.*

   While our current method does not handle missing data, we recognize its importance for future extensions. The main challenge in handling missing values lies in our reliance on kernel-based local averages, which assume regularly spaced observations. We believe that it is possible to modify our framework to handle irregularly spaced observations in the future research. For example, the kernel weights in the local linear estimators could be adapted to reflect the actual spacing between observed points, similar to techniques used in nonparametric regression with irregularly spaced data.

## Reply to referee 1

Thank you very much for the careful reading of our manuscript and the interesting suggestions. In our revision, we have addressed all your comments. Please see our replies to them below.

1. *To address some "big picture" concerns about practical relevance, the authors clarify that rather than the null hypothesis of "no differences ever", the main*



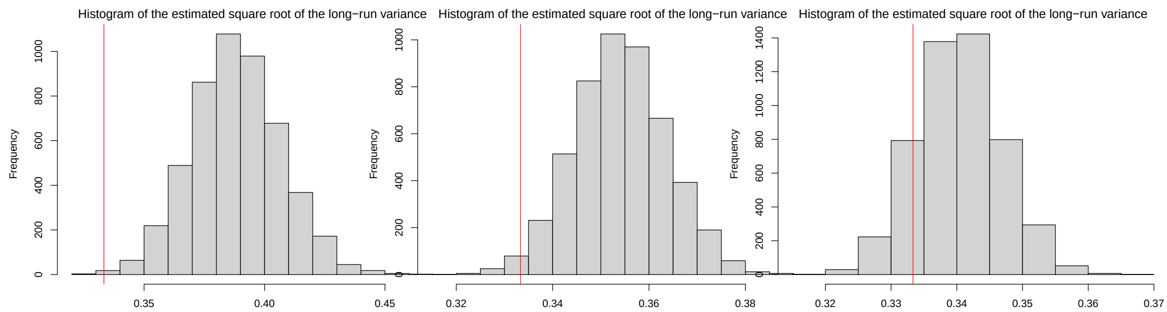

(a) $T = 100$ (b) $T = 250$ (c) $T = 500$

Figure 1: Histogram of the estimated square root of the long-run variance $\sigma$ under the null of $m_i = 0$ for all $i$.

*focus is on the alternative hypotheses, and specifically the ability to detect 1) which series are different and 2) when. This is a reasonable response. However, the simulation studies do not really showcase these priorities. First, simulation results do not appear at all in the main paper. If there is a question about practical relevance of the methods (which the review team did raise), then empirical support should be elevated above some of the technical details. Second, the global size and power tables do not clearly illustrate such local testing capabilities—these metrics would require a much larger collection of data-generating processes to measure performance across a large spectrum of alternative hypotheses. Third, the suggested competitors have not been included, and no compelling reasons were given (more below). Finally, the authors have kept $m_i = 0$ to simulate data under the null "WLOG". As I noted before, this is not WLOG for all methods; that it *is* WLOG for their method suggests that they may observe comparative gains over other approaches for smaller sample sizes (n).*

Although simulation results are primarily in the Supplement due to space constraints, we have highlighted the key findings of the practical relevance in the main text to better showcase our method's local testing capabilities (see p. ...). Marina: I have not done that.

2. *I do not understand why the authors are resistant to add the simple competitors suggested (additive models w/ CI-based testing and Bayesian analogs). Their reasoning was 1) these methods do not offer the same theory guarantees and 2) they include another method, sizer, and a simplified version of their approach. For 1): if the theory actually offers practical gains, then this should show up clearly empirically and further advertise the proposed methods! For 2): sizer is a reasonable competitor (for n=2), but I don't see why that would eliminate consideration of any alternatives. If the question is "which curves are different and when", a natural first approach would be to fit a nonparametric model to each*




(a) $T = 100$ (b) $T = 250$ (c) $T = 500$

Figure 2: Histogram of the estimated square root of the long-run variance $\sigma$ under the null of $m_i$ being the bump function.

*curve and see where/when the confidence/credible intervals do not overlap (w/ or w/o a Bonferroni correction). Beating this approach will be more convincing for general readers.*

We have added the following naive competitor that is based on the uniform confidence bands (UCB) based from Kim (2016):

- We consider the model without the fixed effects $\alpha_i$ since the specification in Kim (2016) does not allow for fixed effects.
- For each of the time series $i = 1, \dots, n$, we calculate the UCB as described in Kim (2016) using bandwidth 0.2. Marina: optimal bandwidth using cross-validation was not that easy to calculate and the theoretical optimal bandwidths were $0.4, 0.33, 0.29$ for $T = 100, 250, 500$, respectively.
- For each pairwise comparison $(i, j)$, we consider the null hypothesis rejected if there is $t \in \{1, \dots, T\}$ with a condition that $t/T \in [0.05, 0.95]$ (since the estimation procedure proposed in Kim (2016) has poor performance at the boundaries, we have decided to exclude them from analysis) such that the UCBs for $m_i(t/T)$ and $m_j(t/T)$ do not overlap.
- We do not apply any correction for the multiple testing problem.
- We consider $\alpha = 0.05$.
- We calculate how many times we reject the null for this naive competitor. The results are presented in Table 8.

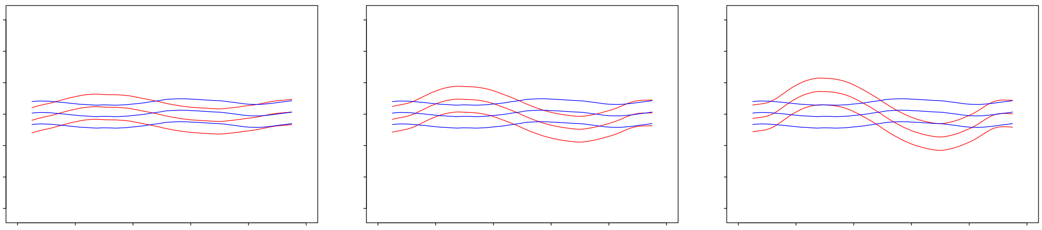
For the smallest sampel size $T = 100$, the UCBs for two trend functions remain relatively wide compared to the strength of the signal so that we almost never detect any differences. For larger sample sizes ($T = 250, 500$), this naive compatitor is able to detect the differences for the strong sygnal. However, we should

| | different bump height $b$ | | | |
|---|---|---|---|---|
| $T$ | 0 | 0.25 | 0.5 | 0.75 |
| $T = 100$ | 0.000 | 0.000 | 0.002 | 0.010 |
| $T = 250$ | 0.000 | 0.001 | 0.029 | 0.638 |
| $T = 500$ | 0.000 | 0.000 | 0.654 | 1.000 |

Table 8: Empirical size and power of the naive competitor from Kim (2016) for $\alpha = 0.05$ for different time series lengths $T$ and bump heights $b$, where $b = 0$ corresponds to the null and $b = 0.25, 0.5, 0.75$ to three different alternatives.

point out that we did not account for multiple testing here and we expect that with Bonferroni correction, the UCBs will become much wider so that even the strongest will not be detectable.

Examples of two UCBs for $T = 500$ and different values of $b$ are presented in Figure 3.



(a) $b = 0.25$ (b) $b = 0.50$ (c) $b = 0.75$

Figure 3: An example of the uniform confidence bands for two time series $i = 1$ and $j \in \{2, \ldots, 15\}$ for $T = 500$ and $b = 0.25, 0.50, 0.75$.

3. *Stochastic volatility (SV) is widely important in the types of applications that the authors mention. Yet the authors 1) claim that extensions to SV would be challenging, 2) elect not to consider any simulation designs with SV to examine robustness, and 3) claim that SV can be handled with log-transformations (and then using the same mean models as proposed). Perhaps 1) is true, but 2) is important based on the claims in the paper and the relevance for this journal. Finally, 3) is not correct for the most widely-used SV models.*

   In order to better mimic behaviour of the stochastic volatility, we have included size and power simulation results with time-varying error variance. Specifically, we assume that for each $i$, the idiosyncratic error process $\{\varepsilon_{it} : 1 \leq t \leq T\}$ follows the AR(1) model $\varepsilon_{it} = a\varepsilon_{it-1} + \sigma(t/T)\eta_{it}$, where $a = 0.25$, the innovations $\eta_{it}$ are i.i.d. normally distributed with mean 0 and standard deviation 1 and

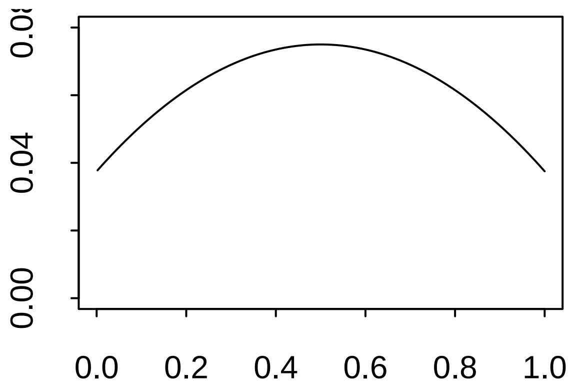



Figure 4: Time-varying error variance $\sigma^2(u)$ for $u \in [0,1]$.

| | nominal size $\alpha$ | | |
|---|---|---|---|
| $T$ | 0.01 | 0.05 | 0.1 |
| 100 | 0.002 | 0.007 | 0.015 |
| 250 | 0.003 | 0.016 | 0.030 |
| 500 | 0.007 | 0.025 | 0.044 |

Table 9: Empirical size of the multiscale test for different time series lengths $T$ and nominal sizes $\alpha$ in case of time-varying error variance.

(a) $b = 0.25$

| | nominal size $\alpha$ | | |
|---|---|---|---|
| $T$ | 0.01 | 0.05 | 0.1 |
| 100 | 0.007 | 0.029 | 0.053 |
| 250 | 0.182 | 0.364 | 0.469 |
| 500 | 0.614 | 0.809 | 0.880 |

(a) $b = 0.5$

| | nominal size $\alpha$ | | |
|---|---|---|---|
| $T$ | 0.01 | 0.05 | 0.1 |
| 100 | 0.008 | 0.035 | 0.061 |
| 250 | 0.850 | 0.957 | 0.980 |
| 500 | 1.000 | 1.000 | 1.000 |

(b) $b = 0.75$

| | nominal size $\alpha$ | | |
|---|---|---|---|
| $T$ | 0.01 | 0.05 | 0.1 |
| 100 | 0.001 | 0.004 | 0.011 |
| 250 | 0.964 | 0.992 | 0.997 |
| 500 | 1.000 | 1.000 | 1.000 |

Table 10: Actual power of the multiscale test for different time series lengths $T$ and nominal sizes $\alpha$ in case of time-varying error variance. Each panel gives the results for a different bump height $b$, where $b = 0.25, 0.5, 0.75$ corresponds to three different alternatives of $m_1$.

$\sigma^2(u) = -0.15(u-0.5)^2 + 0.075$ for $u \in [0,1]$ (see also Figure 4 for $T = 500$). The choice of the coefficients is driven by the requirement $\int_0^1 \sigma^2(u)du = 0.25^2$ that was introduced so that this simulation scenario is comparable to the ones with constant variance. The calculation of the critical values is also performed using the time-varying variance. All of the other parameters of the DGP remain the same. The results are presented in Tables 9 - 12.

| | (a) $b = 0.25$ | | | | (a) $b = 0.5$ | | | | (b) $b = 0.75$ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | nominal size $\alpha$ | | | | nominal size $\alpha$ | | | | nominal size $\alpha$ | | |
| $T$ | 0.01 | 0.05 | 0.1 | $T$ | 0.01 | 0.05 | 0.1 | $T$ | 0.01 | 0.05 | 0.1 |
| 100 | 0.000 | 0.000 | 0.000 | 100 | 0.000 | 0.000 | 0.000 | 100 | 0.000 | 0.000 | 0.000 |
| 250 | 0.004 | 0.017 | 0.030 | 250 | 0.302 | 0.554 | 0.678 | 250 | 0.768 | 0.914 | 0.955 |
| 500 | 0.064 | 0.184 | 0.278 | 500 | 0.997 | 1.000 | 1.000 | 500 | 1.000 | 1.000 | 1.000 |

Table 11: Majority power of the multiscale test for different time series lengths $T$ and nominal sizes $\alpha$ in case of time-varying error variance. Each panel gives the results for a different bump height $b$, where $b = 0.25, 0.5, 0.75$ corresponds to three different alternatives of $m_1$.

| | (a) $b = 0.25$ | | | | (a) $b = 0.5$ | | | | (b) $b = 0.75$ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | nominal size $\alpha$ | | | | nominal size $\alpha$ | | | | nominal size $\alpha$ | | |
| $T$ | 0.01 | 0.05 | 0.1 | $T$ | 0.01 | 0.05 | 0.1 | $T$ | 0.01 | 0.05 | 0.1 |
| 100 | 0.000 | 0.000 | 0.000 | 100 | 0.000 | 0.000 | 0.000 | 100 | 0.000 | 0.000 | 0.000 |
| 250 | 0.000 | 0.000 | 0.000 | 250 | 0.014 | 0.051 | 0.091 | 250 | 0.229 | 0.450 | 0.572 |
| 500 | 0.001 | 0.005 | 0.009 | 500 | 0.757 | 0.919 | 0.956 | 500 | 1.000 | 1.000 | 1.000 |

Table 12: Full power of the multiscale test for different time series lengths $T$ and nominal sizes $\alpha$ in case of time-varying error variance. Each panel gives the results for a different bump height $b$, where $b = 0.25, 0.5, 0.75$ corresponds to three different alternatives of $m_1$.

## Reply to referee 2

Thank you very much for the careful reading of our manuscript and the final acceptance. We have addressed your remaining question in the revision. Please see our reply to it below.

1. *Thank you very much for your very careful revision. I have just one remaining "cosmetic" question: can you explain why in the third row of your figures such as Figure 1, page 27, we see several intervals one op top of the other? In other words is there a meaning of the y-axis of this lower plots, what is its scale? Thank you in advance for adding this information (which I might have missed elsewhere?).*

   The vertical positioning of the intervals from the sets $\mathcal{S}^{[i,j]}(\alpha)$ is purely for visual clarity and does not convey any additional information. We have clarified this in the figure caption.

# References

KIM, K. H. (2016). Inference of the trend in a partially linear model with locally stationary regressors. *Econometric Reviews*, **35** 1194–1220.